

Lesson 12: Traversals Make

45 minutes

Overview

Using Programming Patterns and a step-by-step approach students make their own version of a Random Forecaster app. At the beginning of the lesson students are able to explore a working version of the app. They are then given the design elements of the app but begin with a blank screen. Students use an Activity Guide to go through the high level steps they should use to develop their app but leaves it to them to decide how to write the code. At the end students submit their apps and answer a free-response question in the style of a Create performance task written response prompt. Student apps and written responses can be assessed using a provided rubric.

Standards

Full Course Alignment

CSTA K-12 Computer Science Standards (2017)

- ▶ AP - Algorithms & Programming

Agenda

Warm Up (2 minutes)

Intro the Project

Activity (38 minutes)

Try the Random Forecaster App

Make the Random Forecaster App

Wrap Up (5 minutes)

Assessment: Make Project

Objectives

Students will be able to:

- Implement programming patterns with traversals to develop a functioning app
- Recognize the need for programming patterns with traversals as part of developing a functioning app
- Use debugging skills as part of developing an app
- Write comments to clearly explain both the purpose and function of different segments of code within an app

Preparation

- If time allows, try to build all or part of the Random Forecaster app yourself to understand the challenges involved
- Review the debugging practices you intend to reinforce and resources you'll direct students towards as they get stuck
- Review the Free-Response Supports and consider which, if any, supports are suitable for your classroom.
- Check the "[**Teacher's Lounge**](#)" forum for verified teachers to find additional strategies or resources shared by fellow teachers
- If you are teaching virtually, consider checking our [**Virtual Lesson Modifications**](#)

Links

Heads Up! Please make a copy of any documents you plan to share with students.

For the teachers

- [CSP Unit 6 - Lesson 12 - Traversals Make](#) - Slides [▼ Make a Copy](#)
- [Writing Support - Traversals Make](#) - Handout [▼ Make a Copy](#)

For the students

- [CSP Traversals Make - Random Forecaster App](#) - Activity Guide [▼ Make a Copy](#)

Teaching Guide

Warm Up (2 minutes)

Intro the Project

Remarks

For the past few days, we've learned a lot about using traversals to make apps that manipulate large amounts of information. In today's Make Project you'll be practicing processing lists and using programming patterns with traversals to create a functioning Random Forecaster app.

Teaching Tip

Short Intro: The Warm Up today is short and light. Students should spend the maximum amount of time working on their projects.

Activity (38 minutes)

Try the Random Forecaster App

Group: Make a determination as to whether this project will be completed in pairs or individually. You may even choose to let students decide.

 **Do This:** Have students explore the working Random Forecaster app in Level 1.

 **Discuss:** If students are not working in pairs they should still discuss the prompts with a neighbor.

- *What information is needed to create this app?*
- *What list filtering patterns might be used?*

Discussion Goal:

- What information is needed to create this app?
 - The weather forecast for tomorrow for random cities, including the weather condition, high and low temperatures, and a weather icon.
- What list filtering patterns might be used?
 - The List Filter Pattern: Filtering Multiple Lists is used.

Teaching Tip

Adapting Make Lessons - You may decide that you'd like to have your students write the code for a modified or entirely different app using the same skills. This might help you differentiate your instruction or provide you with a way to reassess students. Check out the [r customizemakelessons_how-to-guide/csp/2025] for guidance on how to get started with this.

Make the Random Forecaster App

Remarks

Now let's build this app. The screen has been set up for you - it's your job to add the code!

Do This: Direct students to level three where they complete the Random Forecaster app. An optional Activity Guide is provided if students would like guidance in creating the app. The most relevant programming pattern is displayed on a slide. Review this pattern quickly with students, if needed.

Submit: Encourage students to check the rubric on the last page of the Activity Guide before submitting.

Teaching Tip

Supporting Students: While students are working on their apps, circulate the room and check in with students who need a little help. Encourage students to collaborate and discuss bugs with each other.

Debugging: Review with students steps they can use to debug if they get stuck:

- Run the code on turtle mode
- Add the variables to the watcher
- Explain the code to a friend

Wrap Up (5 minutes)

Remarks

Great work on your Random Forecaster app today. Before wrapping up, we will practice writing for the style of questions that are asked as part of the Create performance task on the AP exam. Respond to the question in Step 5 in your activity guide, and submit it along with your project.

Do This: Complete the written response in Step 5 of the Activity Guide.

Teaching Tip

Each student, in every classroom, will come to the written response with a different level of comfort, ability, and familiarity with writing for these style of questions. As such, consider offering one or more of the following supports:

Writing Support: Provide students with these sentence starters for guiding their response.

App Interface Support: For students who did not fully complete their app, provide them with this snippet from the exemplar code.

Assessment: Make Project

Use the rubric provided with the project to assess student projects.

✓ Assessment Opportunity ▲

Student responses can be used as a formative assessment to evaluate their understanding of connections between user interface features and functionality. Review their explanations for clarity, accuracy, and alignment with the AP exam's expectations. The provided rubric can be used to guide this review, and to help provide students with feedback to guide their learning and improvement.



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